💛 Check-in Buddy

A gentle daily voice check-in for elderly adults, powered by Alexa+

Hackathon Submission | Team [Your Team Name]

💡 The Problem

Over 50 million adults aged 60 and above live alone in India. Globally, that number exceeds 200 million. Their families are not worried about emergencies. They worry about the small, invisible things: a missed pill, a skipped meal, a voice that sounds quieter than usual.

The solutions available today do not serve this group well:

- Reminder apps rely on screens and rigid schedules. Many older adults do not use smartphones.
- A daily phone call is not always possible for families living across time zones.
- Wearables are expensive, often intrusive, and frequently abandoned after a few weeks.

Voice is the most natural interface for this age group. What if an always-on assistant could hold a real, caring conversation every day?

✅ Our Solution: Check-in Buddy

Check-in Buddy is a simulated Alexa+ experience that acts as a gentle daily wellness companion:

- 🗣️ Holds a short daily voice conversation about medication, meals, mood, and activity.
- 🧠 Remembers previous days and brings them up naturally ("You mentioned your knee hurt yesterday...").
- 🔧 Takes real actions through tool calling: logs medication, meals, and mood; sets reminders; notifies family.
- 🚨 Applies smart escalation rules so families only get alerts when something truly matters.
- 📊 Provides a family dashboard with daily summaries, wellness trends, and alert history.

Why Alexa+? It offers always-on ambient presence with no screen required, high device penetration in homes, and a trusted voice for older adults. This is not an app. It is a companion that is simply there.

💬 Sample Conversation

A typical morning check-in (four turns):

Buddy: Good morning, Amma! Yesterday you said your knee was a bit stiff. How is it feeling today?

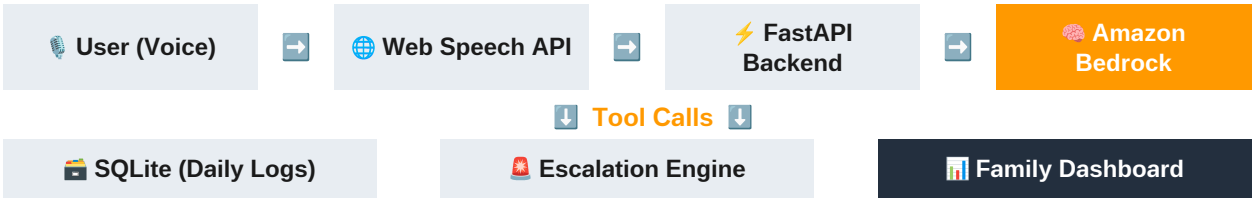
User: *"A little better. I did the stretches the doctor showed me."*

Buddy: That is wonderful! I have logged that. Now, did you take your morning blood pressure tablet? ⚖️ [tool: log_medication]

User: *"Oh... I forgot again. I will take it now."*

Buddy: No worries, taking it now is great! I have set a gentle reminder for tomorrow at 8 AM. Also, this is the second day in a row, so I am letting Priya know just so she is aware. 📩 [tool: notify_family, severity: MEDIUM]

🏗️ Architecture



🚨 Escalation Rules

Smart alerts that filter the noise. Families only hear what matters:

Trigger	Action	Severity
Missed medication two days in a row	Notify family member via push or SMS	⚠️ Medium
"I feel dizzy" combined with a missed meal	Notify family and suggest calling the doctor	🔴 High
Mood declining for three or more consecutive days	Trend alert shown on family dashboard	🟡 Low
No response to check-in for two or more days	Urgent alert and suggest a wellness check	🔴 Critical

🛠️ Tech Stack

Component	Technology
Frontend	HTML and JavaScript with Web Speech API (voice input and spoken replies, plus text fallback)
Backend	Python with FastAPI
AI Engine	Amazon Bedrock (Claude or Nova) with structured tool calling
Memory and Data	SQLite for daily logs, conversation history, and escalation state
Escalation	Rule based engine running over logged data with configurable thresholds
Evaluation	Pytest suite with 50+ scripted conversation scenarios

📊 Results and Metrics

90%+	80%+	10+	100%
Tool Call Accuracy	Escalation Accuracy	Test Scenarios	Safety Guardrails Held

✅ End to end flow working: voice check-in to tool logging to escalation to family alert to dashboard.

⚠️ *Disclaimer: This is a wellness check-in tool, not a medical device. It never diagnoses or gives medical advice. All demo data is synthetic.*

💡 What We Learned

- Evaluation matters just as much as the demo. Scripted scenarios caught tool call edge cases that live testing never would.
- Memory transforms a conversation from scripted to personal. Referencing what happened yesterday made test users visibly more engaged.

- Designing for older adults means pacing, simple language, and patience. The assistant must never rush.
- There is a hard line between helpful check-ins and overstepping into medical territory. We built explicit guardrails for it.

🚀 What's Next

- Real Alexa+ integration when an appropriate development path becomes available.
- Multilingual support, starting with Hindi and Kannada. This is critical for reaching India's elderly population.
- Richer signals such as changes in speech patterns or routine (with user consent) as early wellness indicators.
- Caregiver customization: let families choose which alerts to receive, set thresholds, and manage multiple care recipients.
- A formal review of privacy, consent, and safety with geriatric care and accessibility experts before any real world deployment.

Built with ❤️ for the people who raised us | Open source | Runnable in under 5 minutes